# Inpatient claim review assistant

### The system in one sentence

Help claims staff check inpatient cover and charges, with proof for each recommendation.

### Key design calls

Staff decide every claim. Rejected: Automatic settlement. Why: Rules need insurer approval.

Code sets eligible charges. Rejected: Model maths. Why: Bills need exact checks.

Use the right policy edition. Rejected: Newest upload. Why: Cover changes over time.

Check the saved review. Rejected: Blind retry. Why: Avoid duplicate records.

### Who acts and who decides

| Action | Autonomy |
|---|---|
| Read and cite documents | suggests |
| Check policy charges | acts alone on approved rules |
| Save and assign a draft | acts alone after access checks |
| Approve, reject or pay | human decides |

### Top risks by RPN

FM3 | Room limit and maths error | RPN 225

FM2 | Wrong policy edition | RPN 180

FM4 | Injected or contradictory evidence | RPN 160

FM6 | Unknown save or false completion | RPN 160

FM7 | Review queue stalls | RPN 160

### Launch blockers and owners

H1, H5: First release creates review packets...; Review queue acknowledges assignment within... Owner: Claims lead.

H2: Insurer supplies immutable policy editions... Owner: Policy owner.

H3: Adapters support tenant-scoped reads and... Owner: Platform owner.

H4: Medical documents stay in an... Owner: Privacy owner.

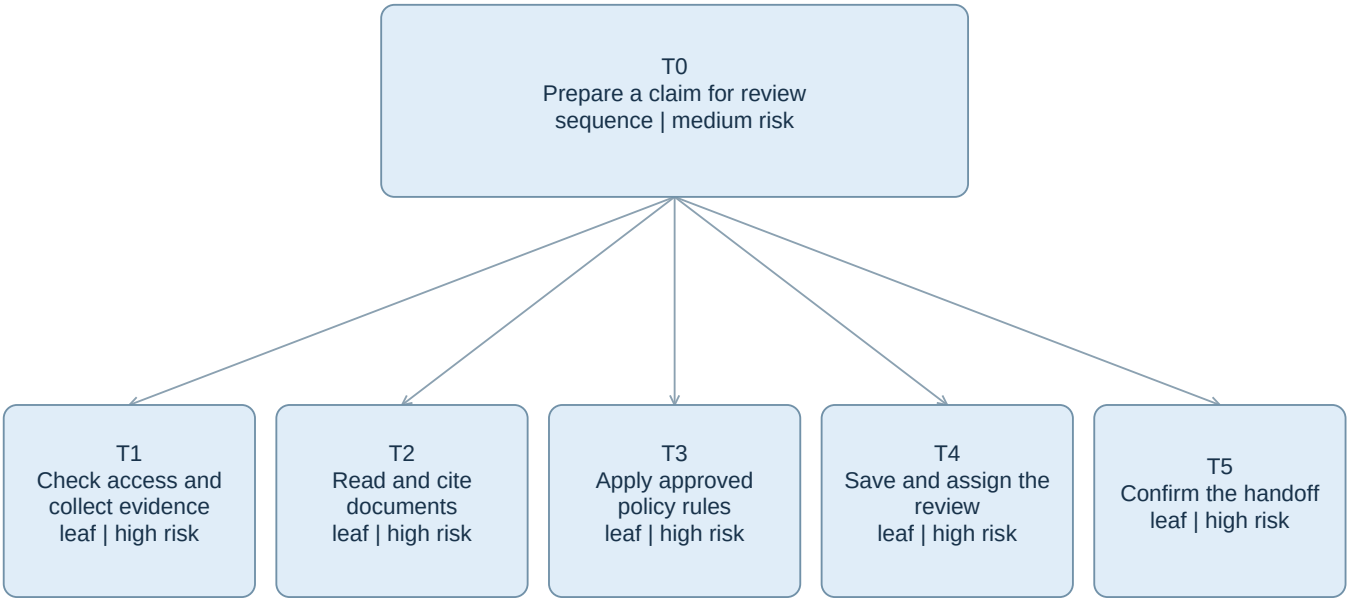
### Illustrative cost per case

USD 0.03780 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 3; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

## Task tree



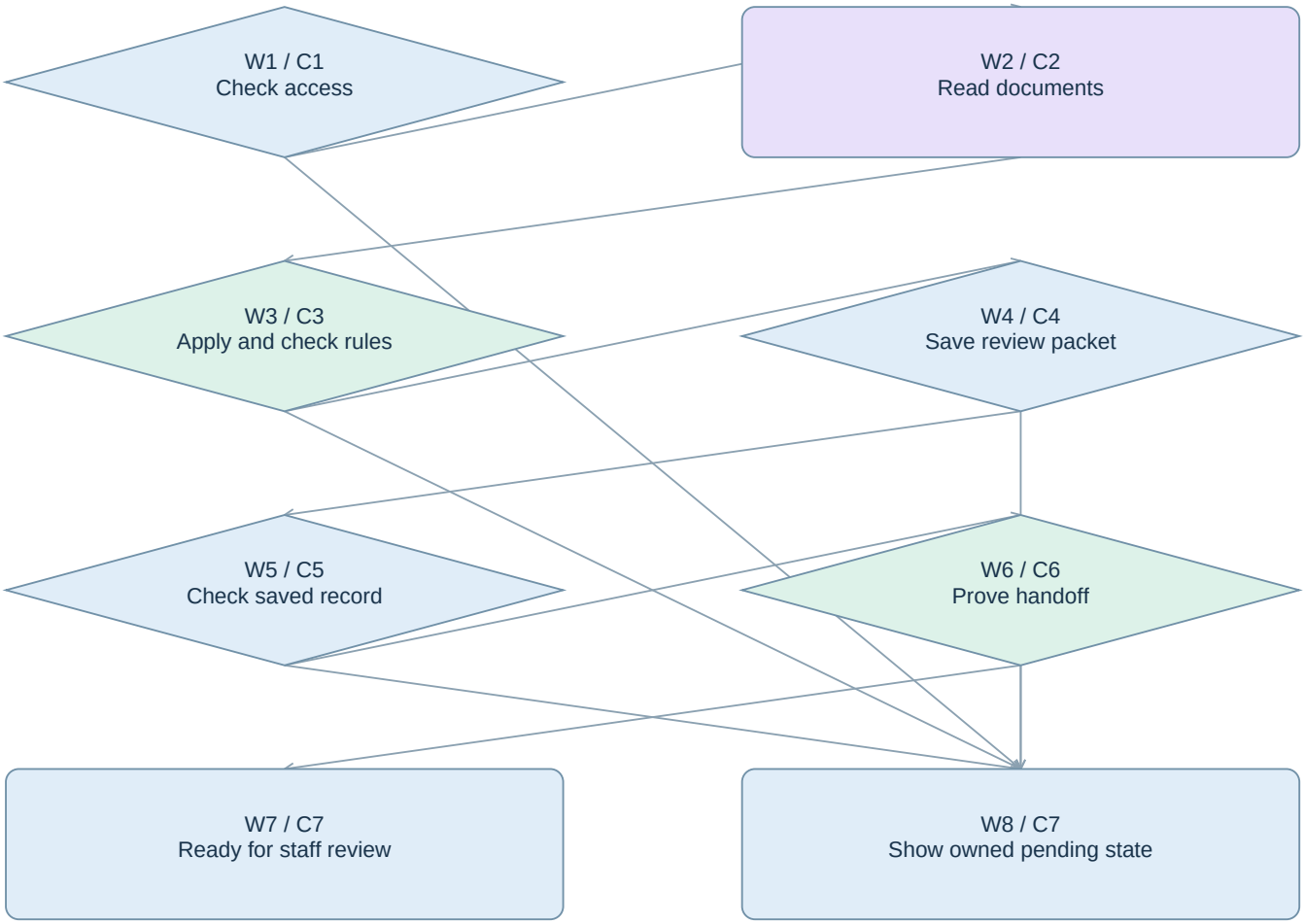

Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

## Plain words and scope

First release prepares and assigns reviews. Staff resolve all claims. Missing facts stay pending. No medical advice is given.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

## Workflow diagram




Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

| Colour | Role |
|---|---|
| #E8E0FA | Agents (LLM) |
| #E0EDF8 | Code: tools |
| #DDF2E9 | Code: validation |
| #FFF0D5 | Code: human triggers |

## Top risks and release checks

## Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

| S / O | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | - | - | FM1 | - | - | - | - | - | - | - |
| 9 | - | - | - | FM2 | FM3 | - | - | - | - | - |
| 8 | - | - | - | FM5, FM6, FM7 | FM4 | - | - | - | - | - |
| 7 | - | - | - | - | - | - | - | - | - | - |
| 6 | - | - | - | - | - | - | - | - | - | - |
| 5 | - | - | - | - | - | - | - | - | - | - |
| 4 | - | - | - | - | - | - | - | - | - | - |
| 3 | - | - | - | - | - | - | - | - | - | - |
| 2 | - | - | - | - | - | - | - | - | - | - |
| 1 | - | - | - | - | - | - | - | - | - | - |

| Risk | Control summary / full register linked |
|---|---|
| FM1 Wrong person or privacy leak | Authenticate before fetch; subject-match pages; regional redaction; no text logs. |
| FM2 Wrong policy edition | Versioned registry date selection; code rechecks versions; missing or conflicting edition means pending. |
| FM3 Room limit and maths error | Explicit charge-class rules and integer minor-unit arithmetic; missing rules means review. Approved formula ID, line class and once-only floor proration; unsupported... |
| FM4 Injected or contradictory evidence | Treat all pages as data; citations/schema; model facts never trusted; ambiguity routes staff. Visible-page OCR only; exact source span and applicable-clause... |
| FM5 Duplicate or stale save | Server conditional versions and unique operation key; no final financial effects. |
| FM6 Unknown save or false completion | Zero write retry; bounded readback, body comparison, record proof, durable on-call recovery. |
| FM7 Review queue stalls | Atomic assigned reviewer required; queue SLA and business-calendar alerts; never mark claim approved. |

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

## Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

| Component / group | Responsibility | Task IDs |
|---|---|---|
| C1 / Code: tools | Trusted case reader | T1 |
| C2 / Agents (LLM) | Evidence reader | T2 |
| C3 / Code: validation | Rule and packet validator | T3, T4 |
| C4 / Code: tools | Review packet store | T4 |
| C5 / Code: tools | Readback and reconciliation | T4, T5 |
| C6 / Code: validation | Record proof check | T5 |
| C7 / Code: tools | Read-only status view | T5 |

## Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

## Architecture and model budget

### F1 - Code owns the workflow

Durable code state machine plus one bounded document reader A single evidence reader fits judgement; persistence, rule maths and recovery remain code.

Why: Keeps evidence, authority and recovery explicit.

### F2 - Keep long-lived state durable

Database-backed worker with transactional outbox and versioned run state Compare state machine and durable workflow engine on recovery, version migrations, audit and operations; choose code worker for one bounded pipeline. No current vendor capability is asserted.

Why: Keeps evidence, authority and recovery explicit.

### F3 - Limit service permissions

Reader can read one authorized claim; writer can create one draft packet; recovery can only read operation status Fresh role/version check at server write; model has no credentials; assignment is provisional and does not approve claim.

Why: Keeps evidence, authority and recovery explicit.

### F4 - Keep medical records separate

Regional encrypted document store plus relational authoritative policy/claim ledger; retrieval index contains redacted citations only Privacy owner approves access/region/retention. Encrypt with scoped keys; audit record IDs and access actors without medical text. Delete model cache immediately; proposed retention...

Why: Keeps evidence, authority and recovery explicit.

### Model cost

Illustrative model cost is USD 0.03780 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

## Evaluation coverage matrix

| Risk / task | EV1 | EV2 | EV3 | EV4 | EV5 | EV6 | EV7 | EV8 | EV9 | EV10 | EV11 | EV12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | X | - | - | - | - | - | - | - | - | X | - |
| FM2 | - | - | X | - | - | - | - | - | - | - | - | X |
| FM3 | X | - | - | X | - | - | - | - | - | - | - | - |
| FM4 | - | - | - | - | X | - | - | - | - | - | - | - |
| FM5 | - | - | - | - | - | X | X | - | - | - | - | X |
| FM6 | - | - | - | - | - | - | - | X | X | X | - | - |
| FM7 | - | - | - | - | - | - | - | - | - | - | - | - |
| T1 | X | X | - | - | - | - | - | - | - | - | X | - |
| T2 | X | - | - | - | X | - | - | - | - | - | X | - |
| T3 | X | - | X | X | X | - | - | - | - | - | - | X |
| T4 | X | - | - | - | - | X | X | X | - | X | - | X |
| T5 | X | - | - | - | - | - | - | X | X | X | - | - |

## Evaluation coverage matrix - continued

| Risk / task | EV13 | EV14 | EV15 | EV16 | EV17 | EV21 | EV22 | EV23 | EV24 | EV25 | EV26 | EV27 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | - | - | - | - | - | - | - | - | - | - | X |
| FM2 | - | - | - | X | X | - | - | - | - | - | - | - |
| FM3 | X | - | - | - | - | X | X | X | - | - | - | - |
| FM4 | - | X | - | X | X | - | - | - | X | X | X | X |
| FM5 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM6 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM7 | - | - | X | - | - | - | - | - | - | - | - | - |
| T1 | - | - | - | - | - | X | X | X | X | X | X | X |
| T2 | - | X | - | X | X | X | X | X | X | X | X | X |
| T3 | X | X | - | X | X | X | X | X | X | X | X | X |
| T4 | - | - | X | X | - | X | X | X | X | X | X | X |
| T5 | - | - | X | - | - | - | - | - | - | - | - | - |

## Evaluation coverage matrix - continued

| Risk / task | EV28 |
|---|---|
| FM1 | - |
| FM2 | - |
| FM3 | X |
| FM4 | - |
| FM5 | X |
| FM6 | - |
| FM7 | - |
| T1 | X |
| T2 | X |
| T3 | X |
| T4 | X |
| T5 | - |

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

## Evaluation plan

| Test | Layer | Purpose |
|---|---|---|
| EV1 | golden | Valid room cap draft |
| EV2 | golden | Cross-tenant rejection |
| EV3 | golden | Historical rule selection |
| EV4 | golden | Missing proration rule routes review |
| EV5 | golden | Injected hospital page |
| EV6 | golden | Concurrent duplicate |
| EV7 | golden | Revoked role at commit |
| EV8 | golden | Save committed but response lost |
| EV9 | golden | Missing proof cannot complete |
| EV10 | golden | Unknown remains owned and frozen |
| EV11 | golden | Raw medical text kept out |
| EV12 | golden | Concurrent policy change |
| EV13 | golden | Minor-unit cap boundary |
| EV14 | golden | Unverified facts and model outage |
| EV15 | golden | Unassigned queue and overdue ownership |
| EV16 | golden | Exclusion is a review candidate |
| EV17 | golden | Policy date boundary conflict |
| EV18 | offline | Held-out evidence and rules |
| EV19 | online | Operational safety monitor |
| EV20 | release | All-case release gate |
| EV21 | golden | Approved linked room proration |
| EV22 | golden | Nonempty class list lacks formula |
| EV23 | golden | Linked class per-line rounding |
| EV24 | golden | Invented source page is blocked |
| EV25 | golden | Wrong policy clause is blocked |
| EV26 | golden | Forged quote is blocked |
| EV27 | golden | Hidden layer never enters evidence reader |
| EV28 | golden | Proposed amount cannot bypass linked code result |

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

## Measurement and rollout

### R1 - Reduce reviewer effort

Baseline then assist. No improvement claim until powered comparison completes.

Why: Keeps evidence, authority and recovery explicit.

### R2 - Protect claimants

All phases. Zero safety incidents; separate strata quality review.

Why: Keeps evidence, authority and recovery explicit.

### R3 - Shadow before staff use

Shadow: proposed 2 weeks minimum. Entry: H2/H4 approved test use and deidentified fixtures. Exit: EV1-EV18 run in harness; SME disagreement resolved; adapter spike H3 complete.

Why: Keeps evidence, authority and recovery explicit.

### R4 - Assist with draft packets

Assist: proposed 2 weeks minimum. Entry: H1-H5 signed and EV20 passes; trained reviewers. Exit: all assignments owned, no safety incident, representative queue SLA evidence and complete metrics. Stop: disable inserts; retain readback/recovery.

Why: Keeps evidence, authority and recovery explicit.

### R5 - Bounded pilot then scale assist

Pilot: randomized assist-only pilot for >=4 weeks after power sizing; scale remains assist-only.. Entry: assist exit, approved sample/power plan and owner authorization. Exit: powered R1 benefit, no safety breach, no worsened audited decision rate and H5 capacity. Scale...

Why: Keeps evidence, authority and recovery explicit.

### R6 - Compare reviewer effort fairly

Pilot. Collect baseline variance and intrareviewer correlation, power a two-sided mean comparison at proposed alpha .05/power .8; adjust reviewer-cluster design effect and attrition. Unknown variance prevents sample claim.

Why: Keeps evidence, authority and recovery explicit.

### R7 - Own failures and recovery

All phases. Signed runbooks, writer kill switch drill, pending-operation recovery drill and queue coverage before launch; only insurer executives may accept residual risk. No acceptance recorded here.

Why: Keeps evidence, authority and recovery explicit.

## Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement only the bounded inpatient review-packet workflow. Preserve evidence gates and draft-only permissions.

## Review dimensions - evidence in review.json

| Dimension | Score | Dimension | Score |
|---|---|---|---|
| D1 | 3/4 | D7 | 3/4 |
| D2 | 3/4 | D8 | 3/4 |
| D3 | 3/4 | D9 | 3/4 |
| D4 | 3/4 | D10 | 3/4 |
| D5 | 3/4 | D11 | 3/4 |
| D6 | 3/4 | D12 | 3/4 |

IR1 major resolved: Added explicit proposed PR1 ratio, class scope, rounding/order and missing-rule fallbacks; EV21-EV23/EV28 specify positive, boundary and prohibited amount branches. Insurer sign-off still required under...

IR2 major resolved: C3 independently checks authorized immutable page, exact quote span and applicable clause before save; EV24-EV26 isolate forged references; C1 visible-only OCR and EV27 exclude...

IR3 minor resolved: Admission July 1 and discharge July 3 now preserve valid chronology while alternate hospital admission June 30 tests the intended source conflict.

## Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.